GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation *12 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● **C.** 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$2.3 \times 10^4 = 0.023 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1}. What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\dfrac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5 + \frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9} + \frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40) + 85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☒ **B.** $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☒ **D.** $\left(\frac{1}{6}\right)^{-1}$
- ☒ **E.** $\sqrt{5^2 + 11}$
- ☒ **F.** $\frac{4^2 + 8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.